Код main.py

```python
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel
from typing import List, Optional
import sqlite3

app = FastAPI(title="Auction API")

# Database setup
def init_db():
    conn = sqlite3.connect('auction.db')
    cursor = conn.cursor()
    cursor.execute('''
    CREATE TABLE IF NOT EXISTS items (
        id INTEGER PRIMARY KEY AUTOINCREMENT,
        name TEXT NOT NULL,
        description TEXT,
        price REAL NOT NULL
    )
    ''')
    conn.commit()
    conn.close()

init_db()


class Item(BaseModel):
    id: Optional[int] = None
    name: str
    description: Optional[str] = None
    price: float


@app.post("/items/", response_model=Item)
def create_item(item: Item):
    conn = sqlite3.connect('auction.db')
    cursor = conn.cursor()
    cursor.execute('''
    INSERT INTO items (name, description, price)
    VALUES (?, ?, ?)
    ''', (item.name, item.description, item.price))
    conn.commit()
    item.id = cursor.lastrowid
    conn.close()
    return item


@app.get("/items/", response_model=List[Item])
def get_items():
    conn = sqlite3.connect('auction.db')
    cursor = conn.cursor()
    cursor.execute('SELECT * FROM items')
    rows = cursor.fetchall()
    conn.close()
    return [Item(id=row[0], name=row[1], description=row[2], price=row[3])
for row in rows]


@app.get("/items/{item_id}", response_model=Item)
def get_item(item_id: int):
    conn = sqlite3.connect('auction.db')
    cursor = conn.cursor()
    cursor.execute('SELECT * FROM items WHERE id = ?', (item_id,))
```

```
    row = cursor.fetchone()
    conn.close()
    if row is None:
        raise HTTPException(status_code=404, detail="Item not found")
    return Item(id=row[0], name=row[1], description=row[2], price=row[3])
```

auction_test.jmx

```
<?xml version="1.0" encoding="UTF-8"?>
<jmeterTestPlan version="1.2" properties="5.0" jmeter="5.6">
  <hashTree>
    <TestPlan guiclass="TestPlanGui" testclass="TestPlan" testname="Auction
API Test Plan" enabled="true">
      <stringProp name="TestPlan.comments">Test plan for FastAPI auction
backend</stringProp>
      <boolProp name="TestPlan.functional_mode">false</boolProp>
      <boolProp name="TestPlan.serialize_threadgroups">false</boolProp>
      <elementProp name="TestPlan.user_defined_variables"
elementType="Arguments" guiclass="ArgumentsPanel" testclass="Arguments"
testname="User Defined Variables" enabled="true">
        <collectionProp name="Arguments.arguments"/>
      </elementProp>
      <stringProp name="TestPlan.user_define_classpath"></stringProp>
    </TestPlan>
    <hashTree>
      <ThreadGroup guiclass="ThreadGroupGui" testclass="ThreadGroup"
testname="Auction Users" enabled="true">
        <stringProp name="ThreadGroup.on_sample_error">continue</stringProp>
        <elementProp name="ThreadGroup.main_controller"
elementType="LoopController" guiclass="LoopControlPanel"
testclass="LoopController" testname="Loop Controller" enabled="true">
          <boolProp name="LoopController.continue_forever">false</boolProp>
          <stringProp name="LoopController.loops">1</stringProp>
        </elementProp>
        <stringProp name="ThreadGroup.num_threads">5</stringProp>
        <stringProp name="ThreadGroup.ramp_time">10</stringProp>
        <boolProp name="ThreadGroup.scheduler">false</boolProp>
        <stringProp name="ThreadGroup.duration"></stringProp>
        <stringProp name="ThreadGroup.delay"></stringProp>
        <boolProp
name="ThreadGroup.same_user_on_next_iteration">true</boolProp>
      </ThreadGroup>
      <hashTree>
        <HeaderManager guiclass="HeaderPanel" testclass="HeaderManager"
testname="HTTP Header Manager" enabled="true">
          <collectionProp name="HeaderManager.headers">
            <elementProp name="" elementType="Header">
              <stringProp name="Header.name">Content-Type</stringProp>
              <stringProp name="Header.value">application/json</stringProp>
            </elementProp>
          </collectionProp>
        </HeaderManager>
        <hashTree/>
        <HTTPSamplerProxy guiclass="HttpTestSampleGui"
testclass="HTTPSamplerProxy" testname="Create Item (POST)" enabled="true">
          <elementProp name="HTTPsampler.Arguments" elementType="Arguments">
            <collectionProp name="Arguments.arguments">
              <elementProp name="" elementType="HTTPArgument">
                <boolProp name="HTTPArgument.always_encode">false</boolProp>
```

```
                <stringProp name="Argument.value">{
  "name": "Item ${__threadNum}",
  "description": "Description for item ${__threadNum}",
  "price": 100.${__Random(10,99,)}
}</stringProp>
                <stringProp name="Argument.metadata">=</stringProp>
              </elementProp>
            </collectionProp>
          </elementProp>
          <stringProp name="HTTPSampler.domain">127.0.0.1</stringProp>
          <stringProp name="HTTPSampler.port">8000</stringProp>
          <stringProp name="HTTPSampler.protocol">http</stringProp>
          <stringProp name="HTTPSampler.path">/items/</stringProp>
          <stringProp name="HTTPSampler.method">POST</stringProp>
          <boolProp name="HTTPSampler.follow_redirects">true</boolProp>
          <boolProp name="HTTPSampler.auto_redirects">false</boolProp>
          <boolProp name="HTTPSampler.use_keepalive">true</boolProp>
          <boolProp name="HTTPSampler.DO_MULTIPART_POST">false</boolProp>
          <stringProp name="HTTPSampler.embedded_url_re"></stringProp>
          <boolProp name="HTTPSampler.postBodyRaw">true</boolProp>
        </HTTPSamplerProxy>
        <hashTree>
          <JSONPostProcessor guiclass="JSONPostProcessorGui"
testclass="JSONPostProcessor" testname="Extract Item ID" enabled="true">
            <stringProp
name="JSONPostProcessor.referenceNames">ITEM_ID</stringProp>
            <stringProp
name="JSONPostProcessor.jsonPathExprs">$.id</stringProp>
            <stringProp name="JSONPostProcessor.match_numbers">1</stringProp>
            <stringProp name="JSONPostProcessor.defaultValues">0</stringProp>
          </JSONPostProcessor>
          <hashTree/>
        </hashTree>
        <HTTPSamplerProxy guiclass="HttpTestSampleGui"
testclass="HTTPSamplerProxy" testname="Get All Items (GET)" enabled="true">
          <elementProp name="HTTPsampler.Arguments" elementType="Arguments">
            <collectionProp name="Arguments.arguments"/>
          </elementProp>
          <stringProp name="HTTPSampler.domain">127.0.0.1</stringProp>
          <stringProp name="HTTPSampler.port">8000</stringProp>
          <stringProp name="HTTPSampler.protocol">http</stringProp>
          <stringProp name="HTTPSampler.path">/items/</stringProp>
          <stringProp name="HTTPSampler.method">GET</stringProp>
          <boolProp name="HTTPSampler.follow_redirects">true</boolProp>
          <boolProp name="HTTPSampler.auto_redirects">false</boolProp>
          <boolProp name="HTTPSampler.use_keepalive">true</boolProp>
          <boolProp name="HTTPSampler.DO_MULTIPART_POST">false</boolProp>
        </HTTPSamplerProxy>
        <hashTree/>
        <HTTPSamplerProxy guiclass="HttpTestSampleGui"
testclass="HTTPSamplerProxy" testname="Get Item By ID (GET)" enabled="true">
          <elementProp name="HTTPsampler.Arguments" elementType="Arguments">
            <collectionProp name="Arguments.arguments"/>
          </elementProp>
          <stringProp name="HTTPSampler.domain">127.0.0.1</stringProp>
          <stringProp name="HTTPSampler.port">8000</stringProp>
          <stringProp name="HTTPSampler.protocol">http</stringProp>
          <stringProp name="HTTPSampler.path">/items/${ITEM_ID}</stringProp>
          <stringProp name="HTTPSampler.method">GET</stringProp>
          <boolProp name="HTTPSampler.follow_redirects">true</boolProp>
          <boolProp name="HTTPSampler.auto_redirects">false</boolProp>
          <boolProp name="HTTPSampler.use_keepalive">true</boolProp>
          <boolProp name="HTTPSampler.DO_MULTIPART_POST">false</boolProp>
        </HTTPSamplerProxy>
```

```
        <hashTree/>
        <ResultCollector guiclass="ViewResultsFullVisualizer"
testclass="ResultCollector" testname="View Results Tree" enabled="true">
          <boolProp name="ResultCollector.error_logging">false</boolProp>
          <objProp>
            <name>saveConfig</name>
            <value class="SampleSaveConfiguration">
              <time>true</time>
              <latency>true</latency>
              <timestamp>true</timestamp>
              <success>true</success>
              <label>true</label>
              <code>true</code>
              <message>true</message>
              <threadName>true</threadName>
              <dataType>true</dataType>
              <encoding>false</encoding>
              <assertions>true</assertions>
              <subresults>true</subresults>
              <responseData>false</responseData>
              <samplerData>false</samplerData>
              <xml>false</xml>
              <fieldNames>true</fieldNames>
              <responseHeaders>false</responseHeaders>
              <requestHeaders>false</requestHeaders>
              <responseDataOnError>false</responseDataOnError>

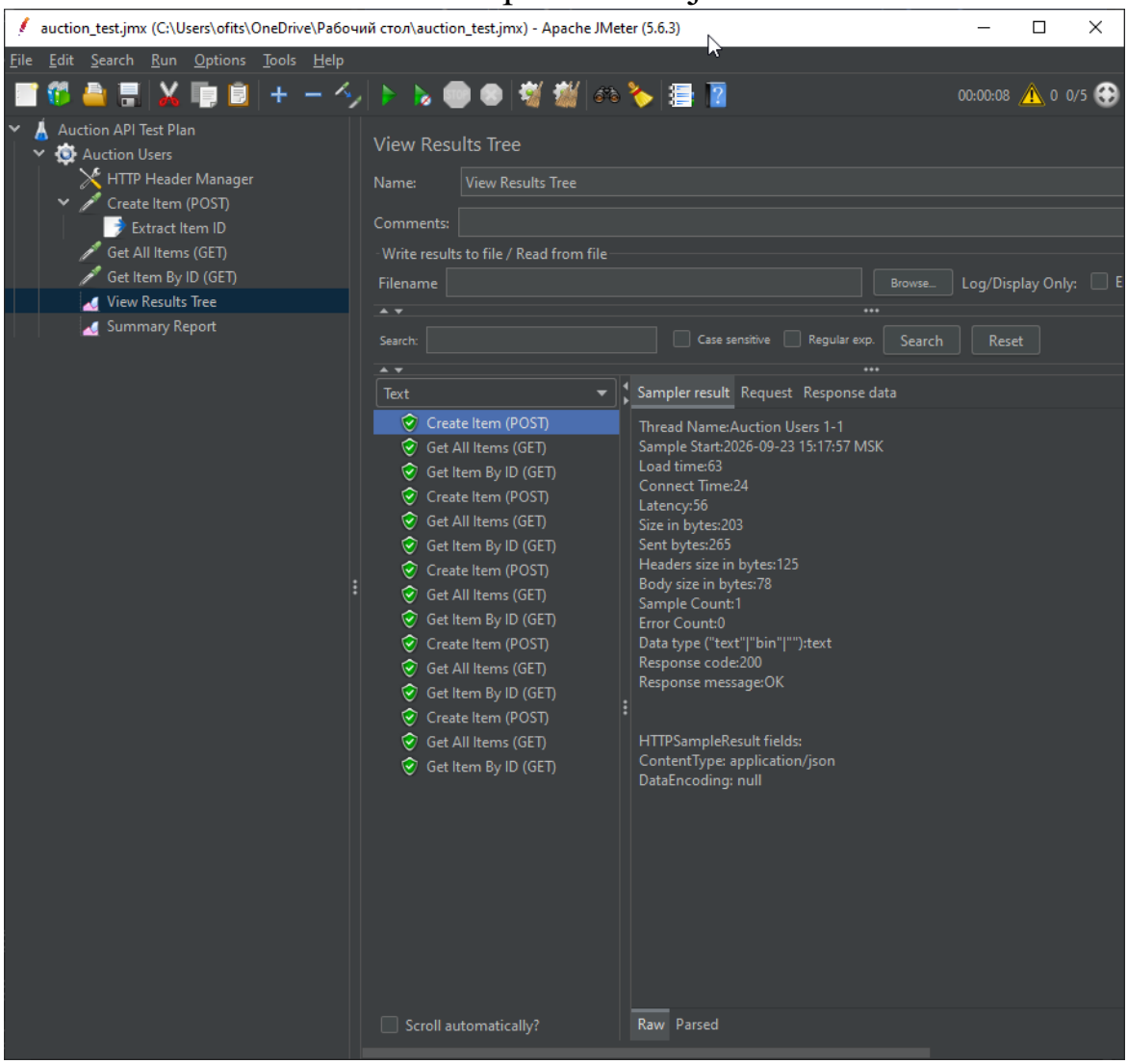
<saveAssertionResultsFailureMessage>true</saveAssertionResultsFailureMessage>
              <bytes>true</bytes>
              <sentBytes>true</sentBytes>
              <url>true</url>
              <threadCounts>true</threadCounts>
              <idleTime>true</idleTime>
              <connectTime>true</connectTime>
            </value>
          </objProp>
          <stringProp name="filename"></stringProp>
        </ResultCollector>
        <hashTree/>
        <ResultCollector guiclass="SummaryReport" testclass="ResultCollector"
testname="Summary Report" enabled="true">
          <boolProp name="ResultCollector.error_logging">false</boolProp>
          <objProp>
            <name>saveConfig</name>
            <value class="SampleSaveConfiguration">
              <time>true</time>
              <latency>true</latency>
              <timestamp>true</timestamp>
              <success>true</success>
              <label>true</label>
              <code>true</code>
              <message>true</message>
              <threadName>true</threadName>
              <dataType>true</dataType>
              <encoding>false</encoding>
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              <subresults>true</subresults>
              <responseData>false</responseData>
              <samplerData>false</samplerData>
              <xml>false</xml>
              <fieldNames>true</fieldNames>
              <responseHeaders>false</responseHeaders>
              <requestHeaders>false</requestHeaders>
              <responseDataOnError>false</responseDataOnError>
```

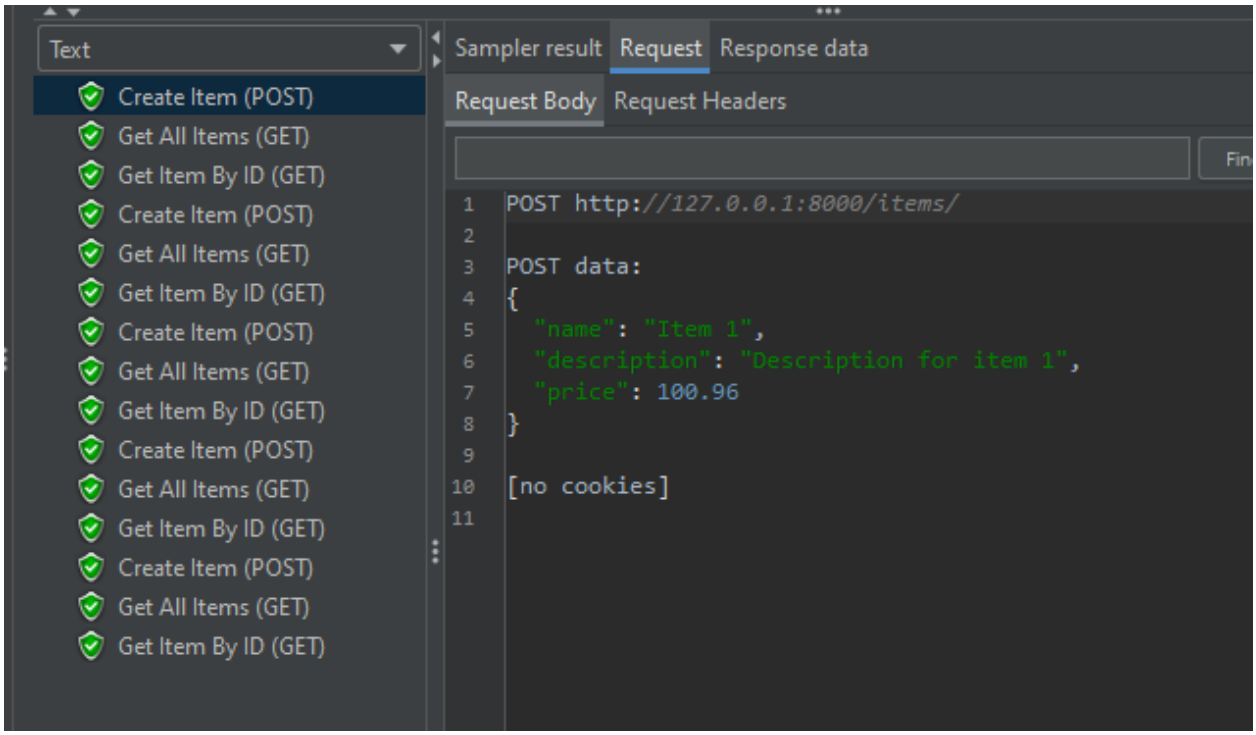
```
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              <bytes>true</bytes>
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              <connectTime>true</connectTime>
            </value>
          </objProp>
          <stringProp name="filename"></stringProp>
        </ResultCollector>
        <hashTree/>
      </hashTree>
    </hashTree>
  </hashTree>
</jmeterTestPlan>
```

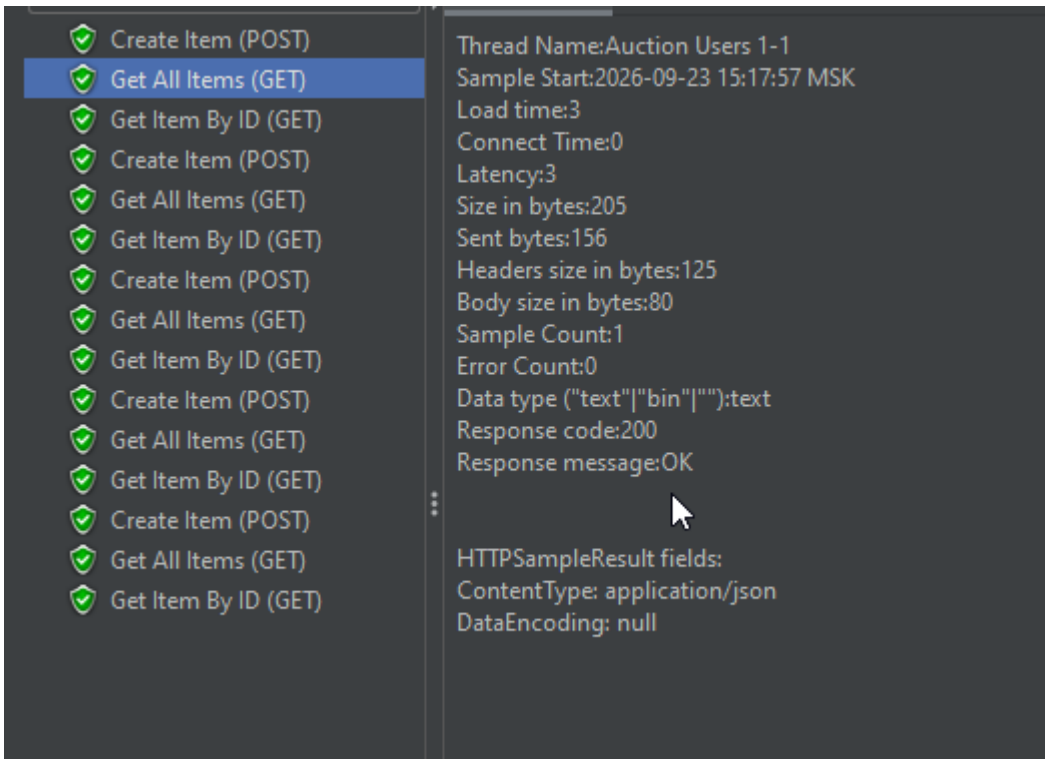
Сервер

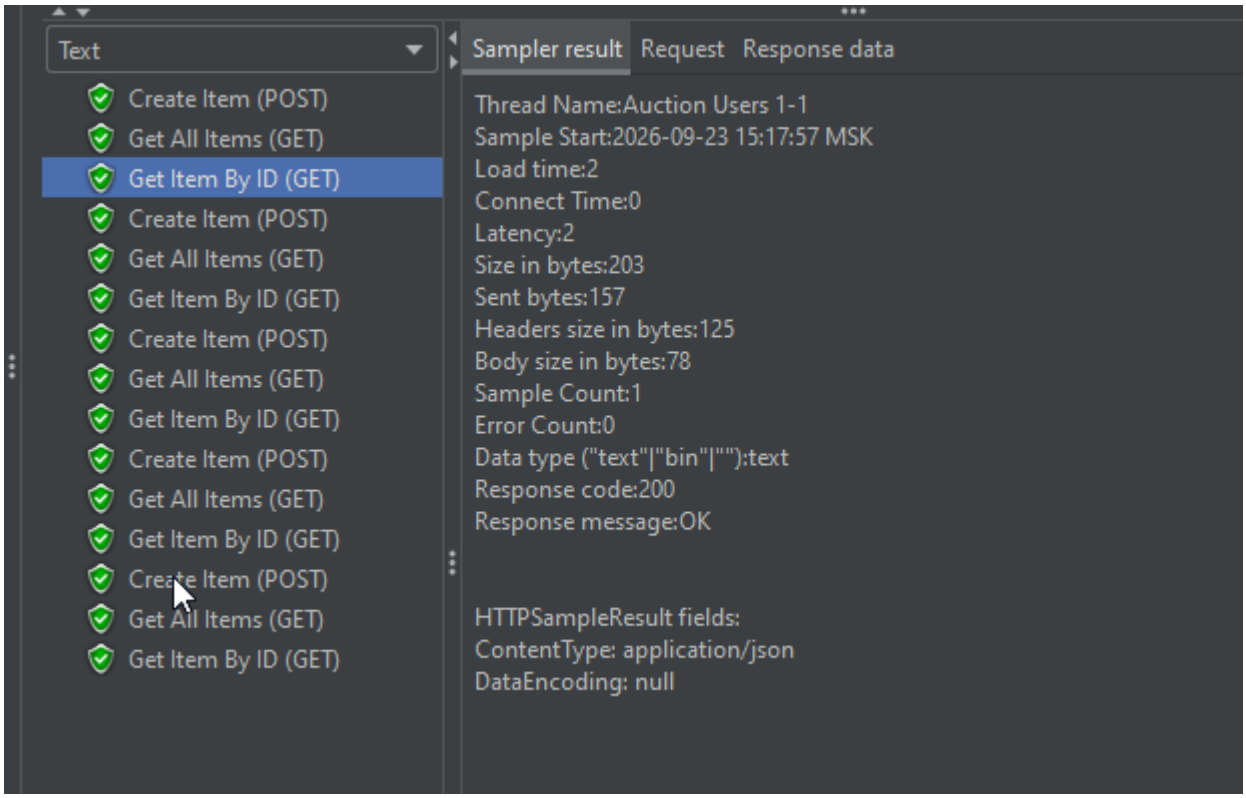
```
(.venv) PS C:\Users\ofits\PycharmProjects\WelcomeScreen>    uvicorn main:app --reload
INFO:     Will watch for changes in these directories: ['C:\\Users\\ofits\\PycharmProjects\\WelcomeScreen']
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [1112] using WatchFiles
INFO:     Started server process [1476]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     127.0.0.1:57121 - "POST /items/ HTTP/1.1" 200 OK
INFO:     127.0.0.1:57121 - "GET /items/ HTTP/1.1" 200 OK
INFO:     127.0.0.1:57121 - "GET /items/1 HTTP/1.1" 200 OK
INFO:     127.0.0.1:57122 - "POST /items/ HTTP/1.1" 200 OK
INFO:     127.0.0.1:57122 - "GET /items/ HTTP/1.1" 200 OK
INFO:     127.0.0.1:57122 - "GET /items/2 HTTP/1.1" 200 OK
INFO:     127.0.0.1:57123 - "POST /items/ HTTP/1.1" 200 OK
INFO:     127.0.0.1:57123 - "GET /items/ HTTP/1.1" 200 OK
INFO:     127.0.0.1:57123 - "GET /items/3 HTTP/1.1" 200 OK
INFO:     127.0.0.1:57124 - "POST /items/ HTTP/1.1" 200 OK
INFO:     127.0.0.1:57124 - "GET /items/ HTTP/1.1" 200 OK
INFO:     127.0.0.1:57124 - "GET /items/4 HTTP/1.1" 200 OK
INFO:     127.0.0.1:57125 - "POST /items/ HTTP/1.1" 200 OK
INFO:     127.0.0.1:57125 - "GET /items/ HTTP/1.1" 200 OK
INFO:     127.0.0.1:57125 - "GET /items/5 HTTP/1.1" 200 OK
```

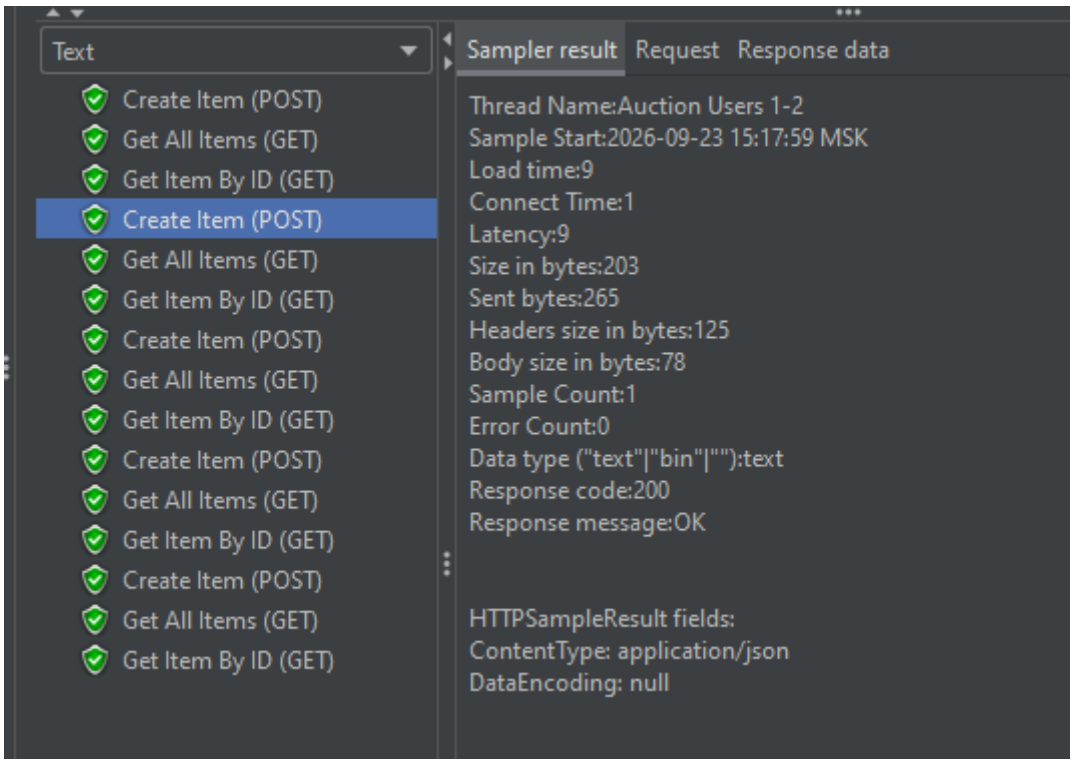
Отображение в jmeter

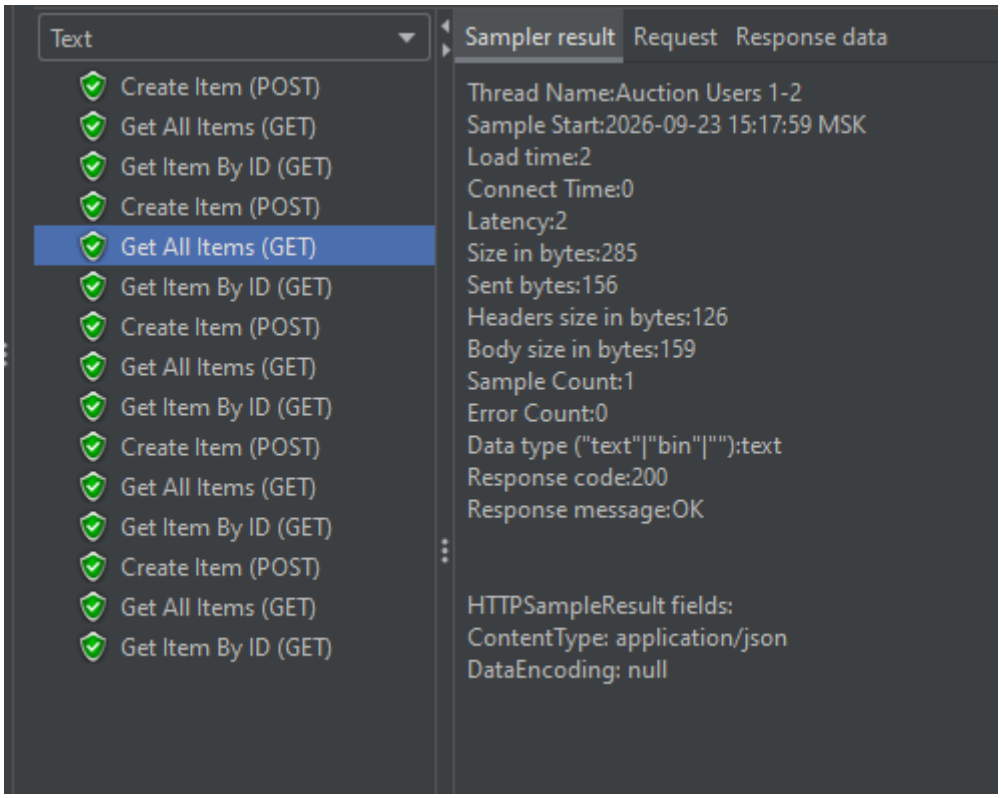


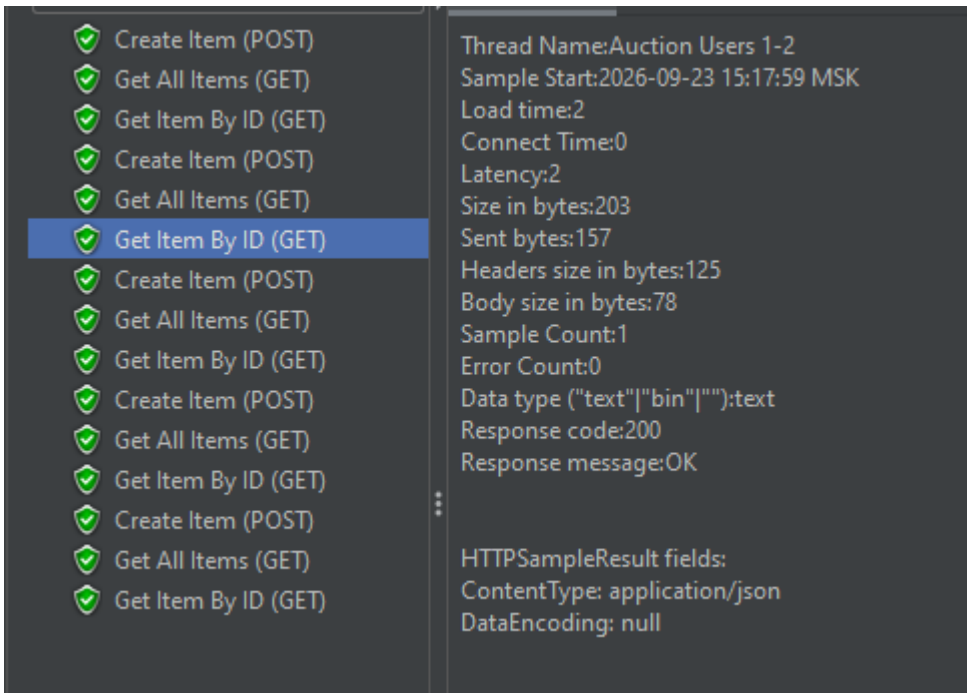


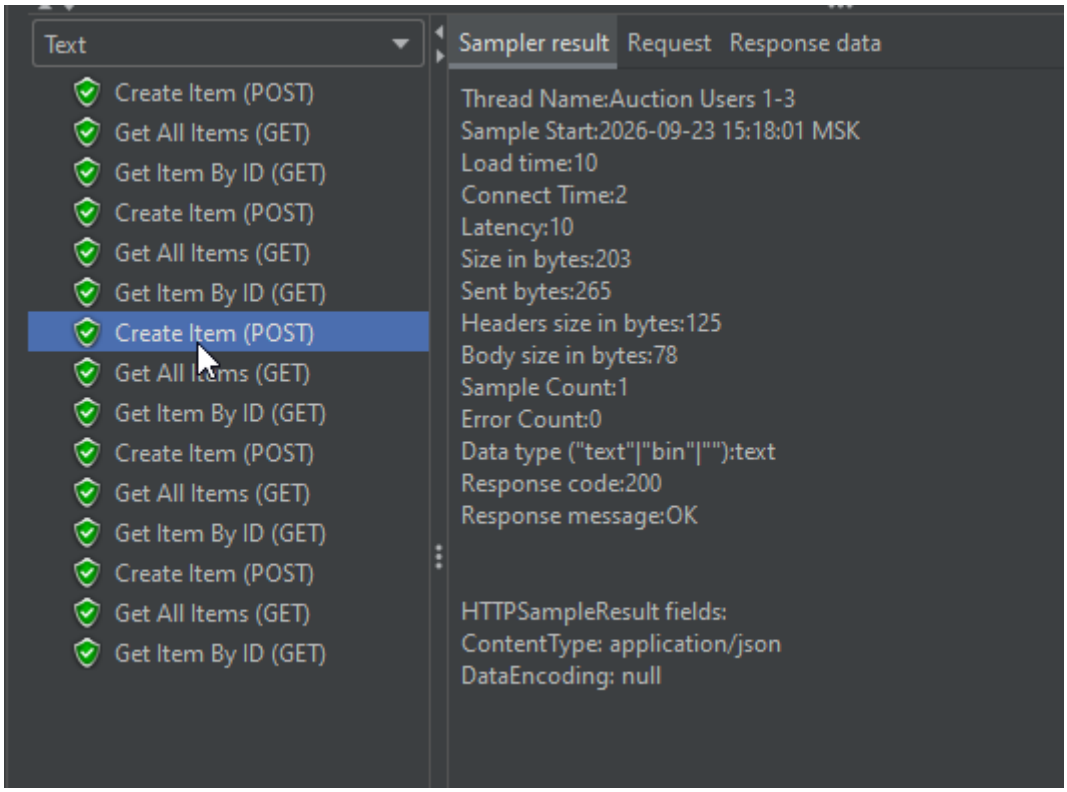

Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Thread Name:Auction Users 1-1
Sample Start:2026-09-23 15:17:57 MSK
Load time:3
Connect Time:0
Latency:3
Size in bytes:205
Sent bytes:156
Headers size in bytes:125
Body size in bytes:80
Sample Count:1
Error Count:0
Data type ("text"|"bin"|""):text
Response code:200
Response message:OK
HTTPSampleResult fields:
ContentType: application/json
DataEncoding: null

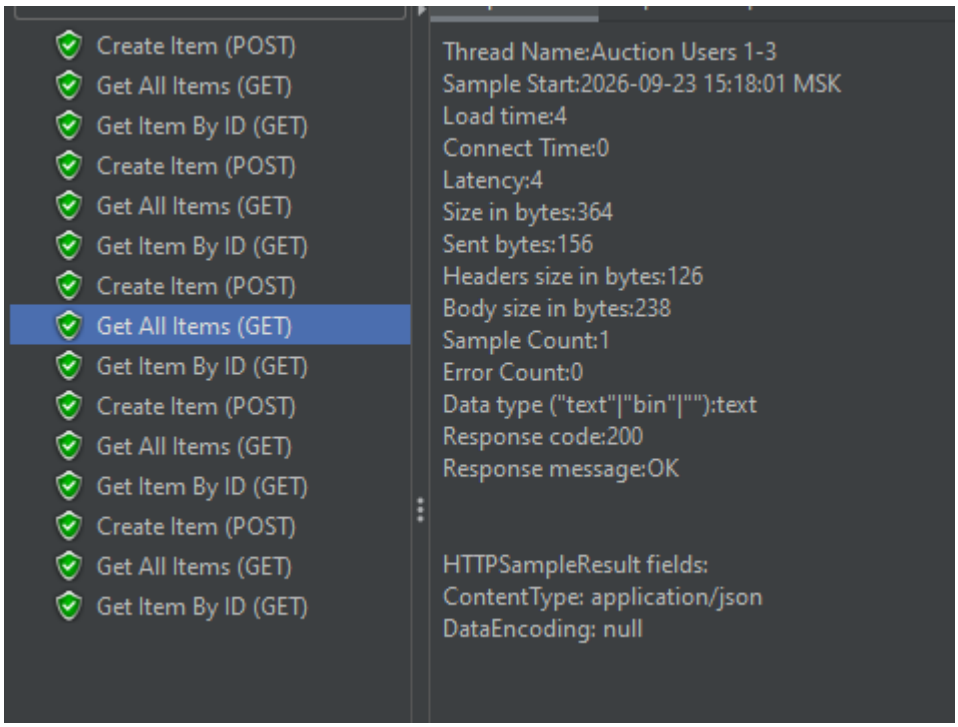

Text
Sampler result
Request
Response data
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Thread Name:Auction Users 1-1
Sample Start:2026-09-23 15:17:57 MSK
Load time:2
Connect Time:0
Latency:2
Size in bytes:203
Sent bytes:157
Headers size in bytes:125
Body size in bytes:78
Sample Count:1
Error Count:0
Data type ("text"|"bin"|""):text
Response code:200
Response message:OK
HTTPSampleResult fields:
ContentType: application/json
DataEncoding: null

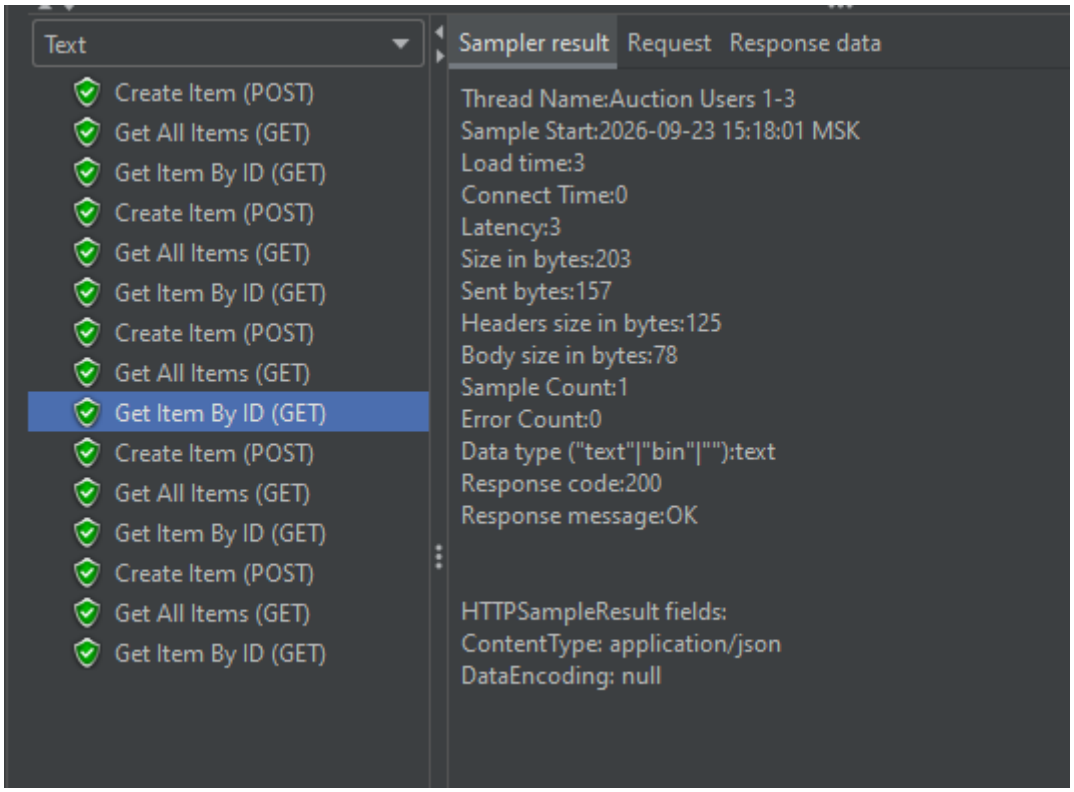

Text
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Sampler result
Request
Response data
Thread Name:Auction Users 1-2
Sample Start:2026-09-23 15:17:59 MSK
Load time:9
Connect Time:1
Latency:9
Size in bytes:203
Sent bytes:265
Headers size in bytes:125
Body size in bytes:78
Sample Count:1
Error Count:0
Data type ("text"|"bin"|""):text
Response code:200
Response message:OK
HTTPSampleResult fields:
ContentType: application/json
DataEncoding: null

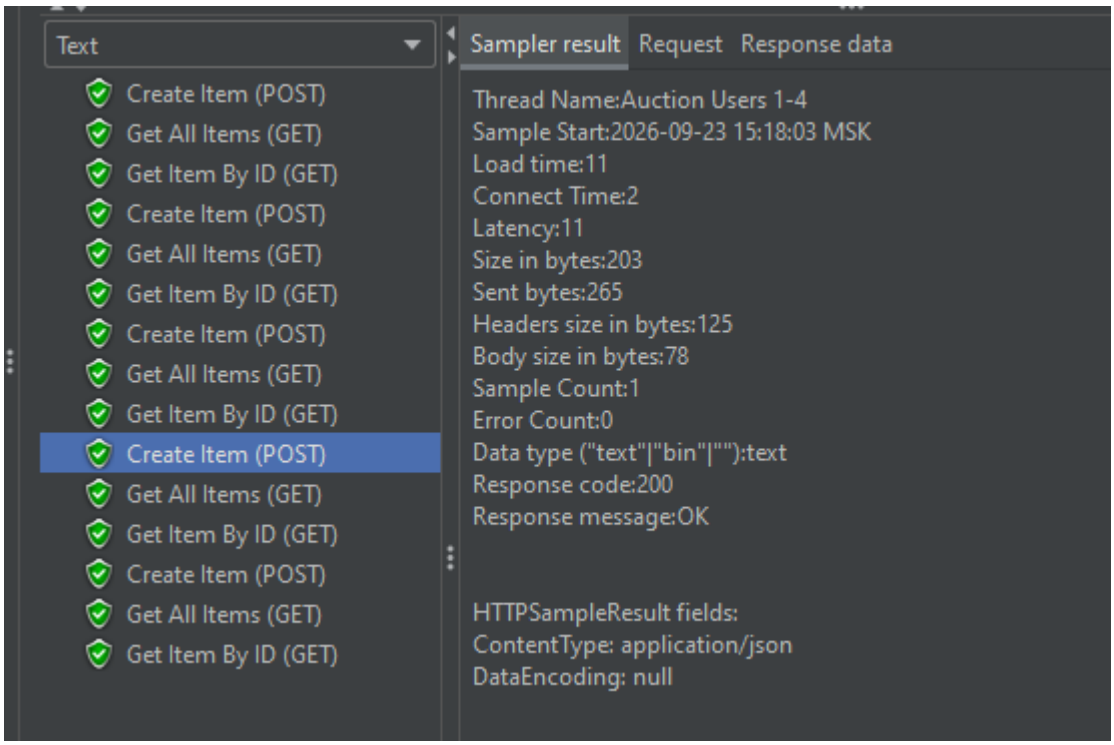

Text
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Sampler result
Request
Response data
Thread Name:Auction Users 1-2
Sample Start:2026-09-23 15:17:59 MSK
Load time:2
Connect Time:0
Latency:2
Size in bytes:285
Sent bytes:156
Headers size in bytes:126
Body size in bytes:159
Sample Count:1
Error Count:0
Data type ("text"|"bin"|""):text
Response code:200
Response message:OK
HTTPSampleResult fields:
ContentType: application/json
DataEncoding: null

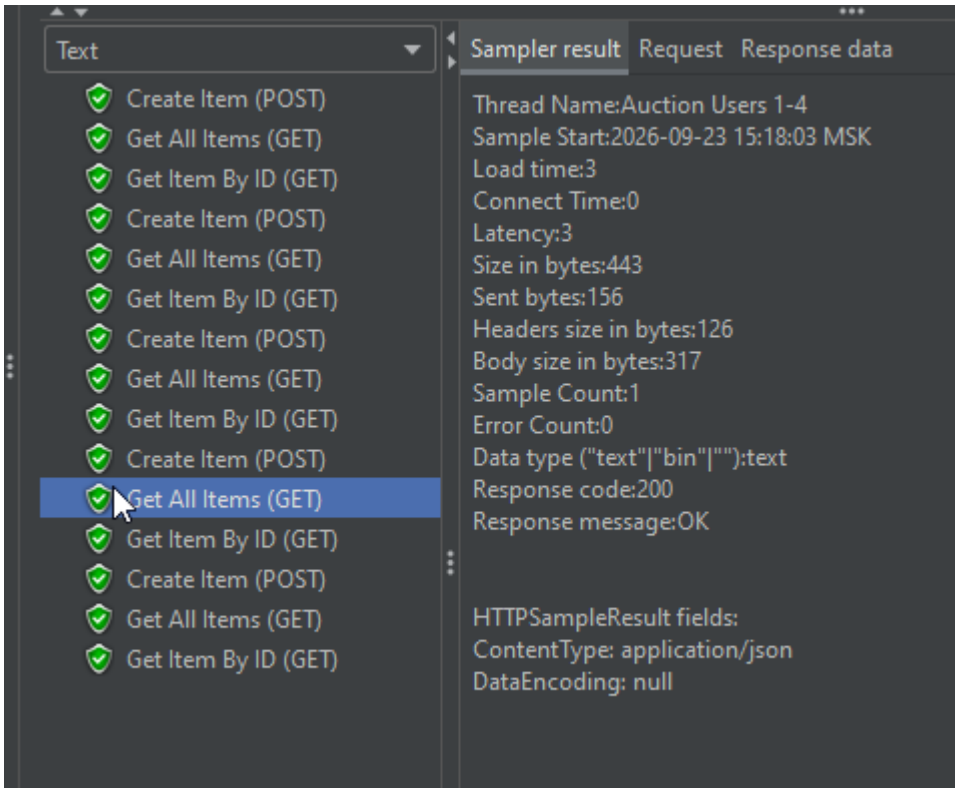

Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Thread Name:Auction Users 1-2
Sample Start:2026-09-23 15:17:59 MSK
Load time:2
Connect Time:0
Latency:2
Size in bytes:203
Sent bytes:157
Headers size in bytes:125
Body size in bytes:78
Sample Count:1
Error Count:0
Data type ("text"|"bin"|""):text
Response code:200
Response message:OK
HTTPSampleResult fields:
ContentType: application/json
DataEncoding: null

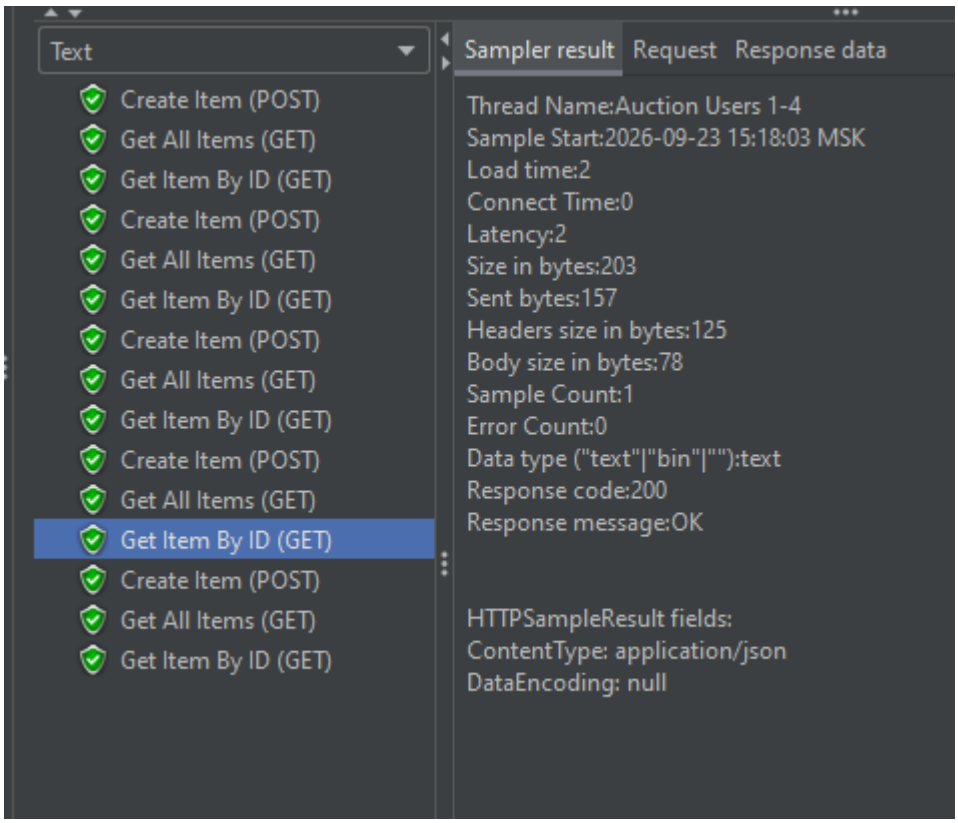

Text
Sampler result
Request
Response data
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Thread Name:Auction Users 1-3
Sample Start:2026-09-23 15:18:01 MSK
Load time:10
Connect Time:2
Latency:10
Size in bytes:203
Sent bytes:265
Headers size in bytes:125
Body size in bytes:78
Sample Count:1
Error Count:0
Data type ("text"|"bin"|""):text
Response code:200
Response message:OK
HTTPSampleResult fields:
ContentType: application/json
DataEncoding: null

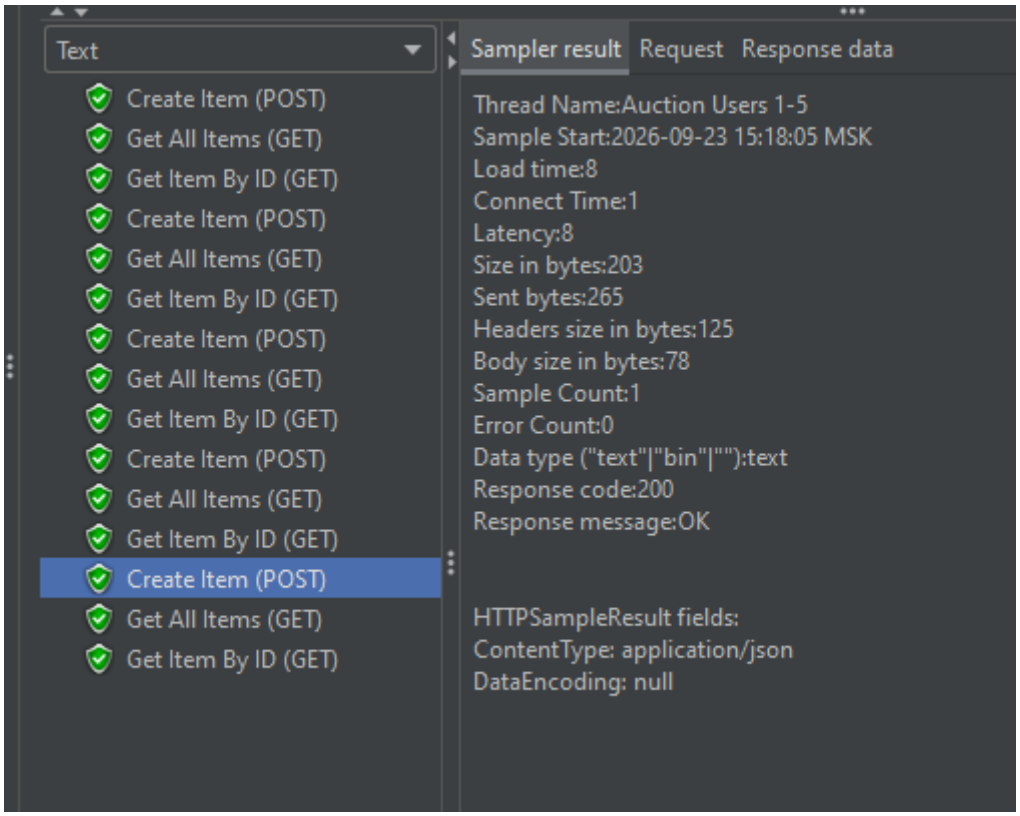

Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Thread Name:Auction Users 1-3
Sample Start:2026-09-23 15:18:01 MSK
Load time:4
Connect Time:0
Latency:4
Size in bytes:364
Sent bytes:156
Headers size in bytes:126
Body size in bytes:238
Sample Count:1
Error Count:0
Data type ("text"|"bin"|""):text
Response code:200
Response message:OK
HTTPSampleResult fields:
ContentType: application/json
DataEncoding: null

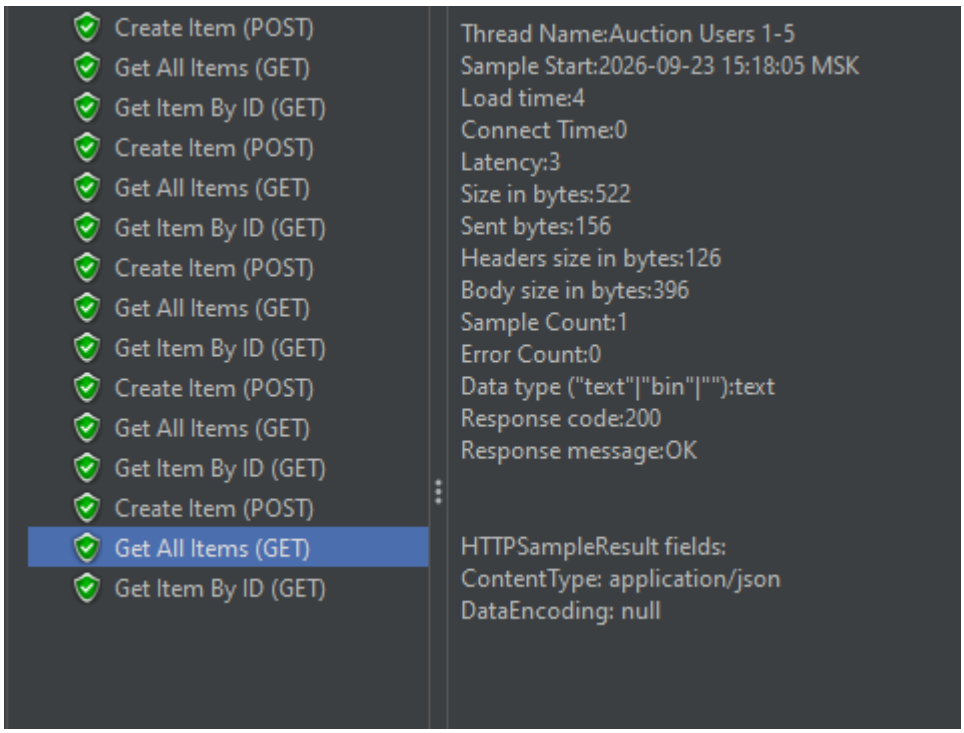

Text
Sampler result
Request
Response data
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Thread Name:Auction Users 1-3
Sample Start:2026-09-23 15:18:01 MSK
Load time:3
Connect Time:0
Latency:3
Size in bytes:203
Sent bytes:157
Headers size in bytes:125
Body size in bytes:78
Sample Count:1
Error Count:0
Data type ("text"|"bin"|""):text
Response code:200
Response message:OK
HTTPSampleResult fields:
ContentType: application/json
DataEncoding: null

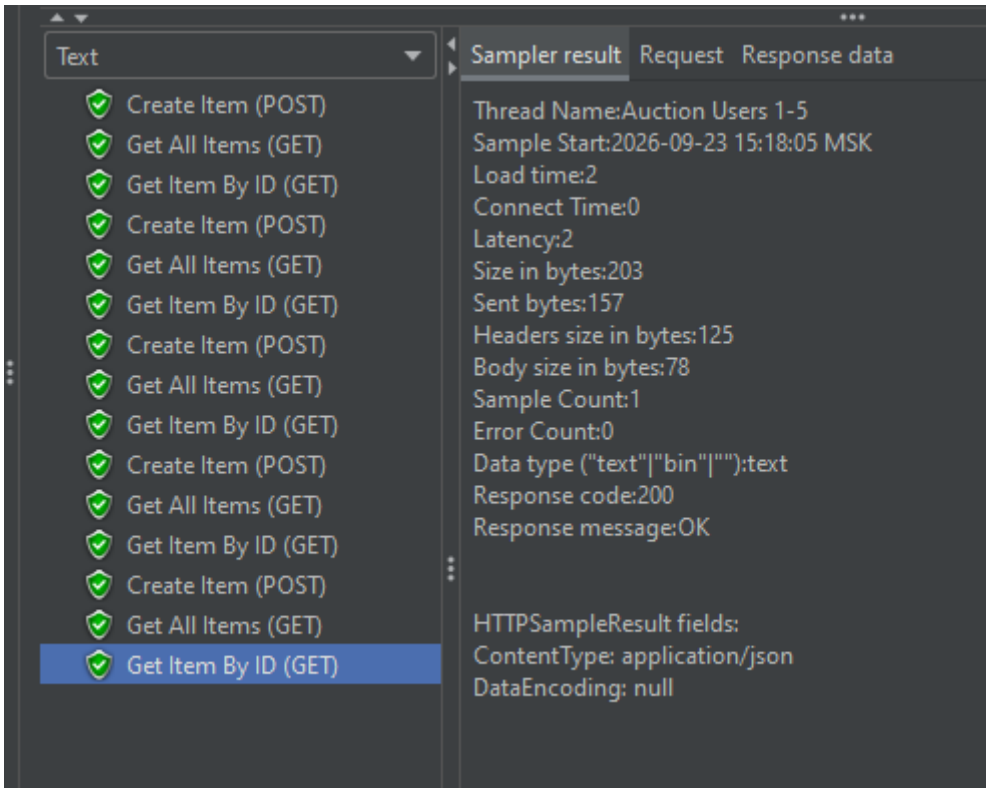

Text
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Sampler result
Request
Response data
Thread Name:Auction Users 1-4
Sample Start:2026-09-23 15:18:03 MSK
Load time:11
Connect Time:2
Latency:11
Size in bytes:203
Sent bytes:265
Headers size in bytes:125
Body size in bytes:78
Sample Count:1
Error Count:0
Data type ("text"|"bin"|""):text
Response code:200
Response message:OK
HTTPSampleResult fields:
ContentType: application/json
DataEncoding: null

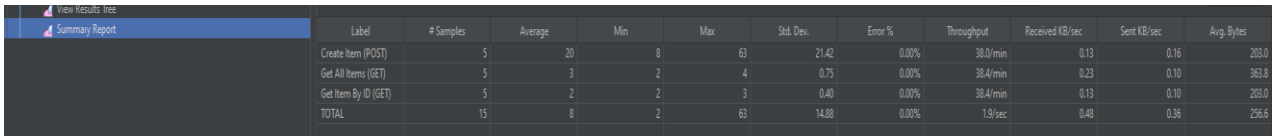

Text
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Sampler result
Request
Response data
Thread Name:Auction Users 1-4
Sample Start:2026-09-23 15:18:03 MSK
Load time:3
Connect Time:0
Latency:3
Size in bytes:443
Sent bytes:156
Headers size in bytes:126
Body size in bytes:317
Sample Count:1
Error Count:0
Data type ("text"|"bin"|""):text
Response code:200
Response message:OK
HTTPSampleResult fields:
ContentType: application/json
DataEncoding: null


Text
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Sampler result
Request
Response data
Thread Name:Auction Users 1-4
Sample Start:2026-09-23 15:18:03 MSK
Load time:2
Connect Time:0
Latency:2
Size in bytes:203
Sent bytes:157
Headers size in bytes:125
Body size in bytes:78
Sample Count:1
Error Count:0
Data type ("text"|"bin"|""):text
Response code:200
Response message:OK
HTTPSampleResult fields:
ContentType: application/json
DataEncoding: null


Text
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Create Item (POST)
Get All Items (GET)
Get Item By ID (GET)
Sampler result
Request
Response data
Thread Name:Auction Users 1-5
Sample Start:2026-09-23 15:18:05 MSK
Load time:8
Connect Time:1
Latency:8
Size in bytes:203
Sent bytes:265
Headers size in bytes:125
Body size in bytes:78
Sample Count:1
Error Count:0
Data type ("text"|"bin"|""):text
Response code:200
Response message:OK
HTTPSampleResult fields:
ContentType: application/json
DataEncoding: null







Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
Create Item (POST)	5	20	8	63	21.42	0.00%	38.0/min	0.13	0.16	203.0
Get All Items (GET)	5	3	2	4	0.75	0.00%	38.4/min	0.23	0.10	363.0
Get Item By ID (GET)	5	2	2	3	0.40	0.00%	38.4/min	0.13	0.10	203.0
TOTAL	15	8	2	63	14.88	0.00%	1.9/sec	0.48	0.36	256.6